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Mechanism and activity of CO oxidation on (001) and (110) surfaces of spinel Co_3O_4, $NiCo_2O_4$ and $NiFe_2O_4$: A DFT + U study

Xiao Shi[a], Steven L. Bernasek[a,b], Annabella Selloni[a,*]

[a] Department of Chemistry, Princeton University, Princeton, NJ 08544, United States
[b] Science Division, Yale-NUS College, 138609, Singapore

A B S T R A C T

We carried out density functional theory (DFT) calculations with on-site Hubbard U repulsion (DFT + U) to investigate the mechanism and activity of the CO oxidation reaction (COOR) on pristine and oxygen-deficient (001) and (110) surfaces of spinel Co_3O_4, $NiCo_2O_4$ and $NiFe_2O_4$. On all investigated (001) surfaces, reaction of adsorbed CO with a surface oxygen atom leads to the spontaneous formation of a CO_2 molecule and a surface oxygen vacancy (V_O). On these surfaces, the COOR is controlled by the thermodynamics of surface re-oxidation through O_2 adsorption at V_O sites. On the Co_3O_4 (110) surface, instead, the kinetic barrier for CO_2 formation is rate determining, while re-oxidation is thermodynamically favorable also at relatively high temperature (T ~ 500 K). In humid environment, water adsorption at surface V_Os inhibits O_2 adsorption thus blocking the reaction. This effect is particularly important on $NiFe_2O_4$(001), which may contribute to the limited COOR activity of this material.

1. Introduction

The discovery that spinel Co_3O_4 nanorods can efficiently oxidize CO to CO_2 at temperatures as low as −70 C [1] has triggered a great deal of interest in the catalytic properties of this material in recent years [2-5]. Interest in Co_3O_4 has also generated efforts aimed at modifying its catalytic activity through doping or substitution with other earth-abundant elements. Among such substituted cobaltites, nickel cobaltite, $NiCo_2O_4$ (NCO), has emerged as a particularly promising catalyst for low-temperature CO and methane oxidation [6,7], and as an electrocatalyst for the oxygen evolution reaction (OER) as well [3,8,9]. A related spinel oxide, nickel ferrite $NiFe_2O_4$ (NFO), is also highly active for methane complete oxidation [10] and the OER [11], suggesting similar reaction mechanisms for these different materials.

Among oxidation reactions, that of CO to CO_2 is of great interest both as the reaction used for reducing the amount of CO in waste gases, and as a benchmark reaction for oxidation catalysts. Numerous experimental studies of the CO oxidation reaction (COOR) on Co_3O_4 have been reported, and the (110) surface has been found to be the most active low-index surface of this material: CO can be converted to CO_2 even at −70 °C, and complete conversion takes place at around 150 °C [1,7]. However, room temperature CO oxidation on Co_3O_4 (110) gradually deactivates during the reaction, which has been attributed to surface reconstruction and water adsorption [12,13]. Moreover, (110) surfaces are not very common for the spinel structure; in particular, (001) surfaces have lower surface energies and thus generally make larger contributions to the surface area [14], but have been reported to be inactive for CO oxidation [1]. On the theoretical side, numerous studies examined CO oxidation on Co_3O_4 (110) and found evidence of a Mars–van Krevelen mechanism [15] with a major role of Co ions at octahedral sites [16–18]. However, the origin of the different activities of the Co_3O_4 (110) and (001) surfaces has not been analyzed in detail.

In this work, our aim is to address this issue and at the same time compare the reactivity of cobalt oxide, nickel cobaltite and nickel ferrite. To this end, we have carried out a computational study of the COOR over pristine and defected Co_3O_4(110), Co_3O_4(001), $NiCo_2O_4$(001) and $NiFe_2O_4$(001) surfaces, using DFT calculations with the addition of on-site Coulomb repulsion U terms on Co, Ni and Fe 3d shells (DFT + U) [19]. Our results suggest different rate determining steps for (110) and (001) surfaces, surface reoxidation and CO_2 formation being rate limiting on (001) and (110) surfaces, respectively.

2. Methods and models

Spin polarized DFT + U calculations were performed using the plane-wave-pseudopotential scheme as implemented in the Quantum Espresso package [20]. Exchange and correlation terms were described using the Perdew-Burke-Ernzerhof (PBE) [21] functional with on-site Coulomb repulsion U term on Co, Ni and Fe 3 d electrons. It is known that the U value can affect calculated properties and no single value of U is equally appropriate for all properties [27]. Here, we used the values U(Co) = 3.0 eV, U(Ni) = 5.5 eV and U(Fe) = 3.5 and 3.4 eV for Fe

* Corresponding author.
E-mail address: aselloni@exchange.Princeton.EDU (A. Selloni).

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at Oh and Td sites that are found to provide a better description of Co_3O_4 and NCO and NFO's thermodynamic properties compared to the U values determined from Linear Response theory [22,23]. Ultrasoft pseudopotentials [24] were employed and the valence electrons include O 2 s, 2 p; Ni 3 d, 4 s; Co 3 d, 4 s and Fe 3 d, 4 s states. Kinetic energy cutoffs used were 35 and 350 Ryd for wave functions and augmented density on Co_3O_4 and NCO, and 50 and 500 Ryd for wave functions and augmented density on NFO. Structural optimizations were carried out by relaxing all atoms until forces were smaller than 1.0×10^{-3} a.u.

Co_3O_4, NCO and NFO (001) surfaces were modeled using slabs of 11 layers terminated by oxygen and metal cations occupying octahedral sites in the bulk material. The Co_3O_4 (110) surface was modeled using slabs of 7 layers with oxygen rich termination. A vacuum region of 20 Å was used to separate adjacent slabs. We kept similar chemical composition for the different (001) slabs, with 10 metal atoms at tetrahedral (Td) sites; 24 metal atoms at octahedral (Oh) sites and 48 oxygens. The Co_3O_4 (110) slab included 14 (6) Co atoms at Oh (Td) sites and 28 oxygens. We employed theoretically optimized lattice parameters, and considered square (2×2) surface unit cells of dimensions 8.136×8.136 Å^2, 8.209×8.209 Å^2 and 8.447×8.447 Å^2 for Co_3O_4, NCO and NFO (001) surfaces, respectively [22, 23, 25]; the unit cell dimensions for Co_3O_4 (110) were 8.149×5.762 Å^2. We sampled the surface Brillouin zone using a ($2 \times 2 \times 1$) k-point grid for both (001) and (110) surfaces. CO and O_2 molecules were adsorbed only on one of the two surfaces of each slab; previous calculations showed that dipole corrections have minor influence (0.01 eV) and were not included in our setup.

Kinetic barriers for selected reaction steps were determined using the Climbing Image Nudged Elastic Band (CI-NEB) method [26]. We typically used 9 images that we optimized until forces were smaller than 0.05 eV/Å. We confirmed the correctness of the transition state structure by verifying that the state can relax to either the initial or final state of the pathway after small distortion.

Adsorption energies of various species X (e.g. CO, O_2, etc) were calculated using

$$E_{ads} = E_{X*} - E_{surf} - \mu_X$$

where E_{X*} and E_{surf} are the computed total energies of the surfaces with and without adsorbed X, respectively, and μ_X denotes the chemical potential of X calculated from JANAF Tables [28]. Specifically, calculations were performed at both T = 0 K and T = 300 K with CO, O_2 and CO_2 partial pressures of 0.67 atm, 0.33 atm and 0.0004 atm, respectively.

3. Results and discussion

3.1. Co_3O_4 (001) and (110) surfaces

CO adsorption energies on Co_3O_4(001) and (110) surfaces are reported in Table 1. For Co_3O_4 (001), both the pristine (defect-free) and partially reduced (with one oxygen vacancy, V_O, per unit cell) surfaces were examined (Fig. 1(a, b)), as the V_O concentration can influence further V_O formation and therefore the energetics of CO oxidation through the Mars–van Krevelen (MvK) mechanism. For Co_3O_4 (110), we considered only the pristine surface (Fig. 1(c, d)), as the defected (110) surface tends to reconstruct [22].

For CO at the 5-fold coordinated Co(Oh) site on Co_3O_4(001), the computed adsorption energy is $E_{ads} = -0.95$ (-0.41) eV at T = 0 (300) K and the adsorbed molecule is perpendicular to the surface with C–Co distance of 1.75 Å, suggesting a rather strong interaction. CO adsorption is less favorable at V_Os, while it is much more favorable at the two nonequivalent O1 and O2 oxygen sites[29], $E_{ads} = -1.86$ (-1.32) and -1.43 (-0.89) eV, respectively (Fig. 1(a)). When CO adsorbs at the latter sites, it actually forms a CO_2 molecule without any apparent intermediate or barrier. The resulting adsorption configuration can be described as a CO_2 at a surface V_O (see structure II in Fig. 2), consistent with a MvK mechanism.

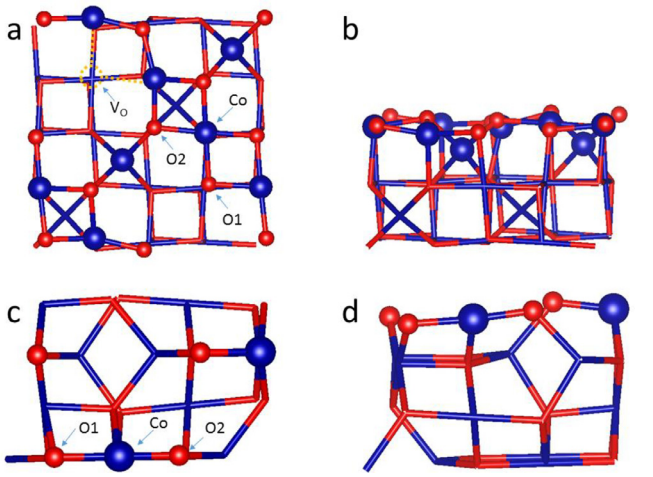


Fig. 1. (**a**), Top view of the (2×2) surface unit cell and (**b**), side view of the top few layers of the Co_3O_4 (001) slab model used in our calculations. The atoms of the top and second layer are shown as balls. The dashed orange circle in **a** indicates a surface oxygen vacancy (V_O); other labels indicate: Co, cobalt at a surface octahedral site; O1, surface oxygen bonding with 3rd layer Co(Oh); O2, surface lattice oxygen bonding with 2nd layer Co(Td). (**c**) Top view of the surface unit cell and **d**, side view of the first few layers of the pristine Co_3O_4 (110) slab model used in our study. Labels indicate: Co, cobalt at a surface octahedral site; O1, surface oxygen bonding with 2 inner layer Co(Oh) and 1 surface Co(Oh); O2, surface oxygen bonding with 1 inner layer Co(Td) and 1 surface Co(Oh).

On Co_3O_4(110), the computed adsorption energy for CO at the 4-coordinated surface Co(Oh), $E_{ads} = -1.80$ (-1.26) eV, is more negative (favorable) than on the (001) surface. CO points toward one of the Co(Oh)-O bonds, forming an angle of 45° with the surface. In addition to Co(Oh), two types of oxygen sites are available for CO adsorption (Fig. 1(c)): O1, bound to three Co(Oh), and O2, bound to one Co(Oh) and one Co(Td). The CO adsorption energy is larger at O2, $E_{ads} = -1.91$ (-1.38) eV vs. $E_{ads} = -1.49$ (-0.96) eV at O1. At both these sites, CO forms bridging bonds with the oxygen and a surface Co (Oh), with bond lengths of 1.30 and 1.95 Å at O1 and O2, respectively (structure II in Fig. 3, and Fig. S1 of Supporting Information).

Our results for the intermediates and energetics of the COOR on Co_3O_4 (001) and Co_3O_4(110) are summarized in Figs. 2, 3 and Table 2. The first three structures in Fig. 2 correspond to the bare (001) surface (**I**), the adsorption of a CO molecule and formation of CO_2 by abstraction of a surface oxygen atom (**II**), and the surface with an oxygen vacancy, after the desorption of CO_2 (**III**). In the subsequent step, an O_2 molecule adsorbs at the V_O site, with one of the oxygen atoms taking

Table 1
Computed CO adsorption energies (in eV) at surface cobalt, oxygen and oxygen vacancy sites on pristine and defected Co_3O_4 (001) and (110) surfaces. Adsorption energies are reported at both T = 0 K and T = 300 K (in parentheses) with CO pressure of 0.67 atm.

	Co	O1	O2	V_O
Pristine 001	−0.95(−0.41)	−1.86(−1.32)	−1.43(−0.89)	
Defected 001	−0.90(−0.37)	−1.75(−1.21)	−1.04(−0.51)	−0.65(−0.12)
Pristine 110	−1.80(−1.27)	−1.49(−0.96)	−1.91(−1.38)	

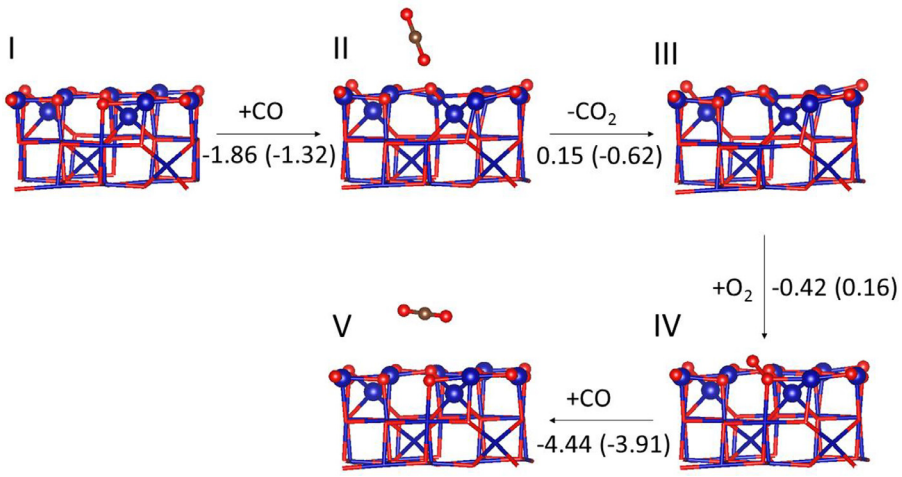


Fig. 2. Intermediates (side views) and free energy changes (in eV) for the COOR on pristine Co_3O_4 (001): pristine surface (I); adsorbed CO, forming a CO_2 molecule with a surface O atom (II); defective surface following CO_2 desorption (III); re-oxidized surface, with an O_2 molecule at the vacancy site (IV); pristine surface with physisorbed CO_2 formed upon adsorption of another CO (V). Free energy changes are given at both T = 0 K and T = 300 K (between parentheses) with CO, O_2 and CO_2 partial pressures of 0.67 atm, 0.33 atm and 0.0004 atm, respectively.

the position of the missing lattice oxygen, while the other oxygen stretches outwards (**IV**); the O—O bond length is 1.344 Å, suggesting a superoxide species, O_2^-. This is also confirmed by the presence of an unpaired spin localized mostly on the outer oxygen. The computed O_2 adsorption energy is −0.42 eV at T = 0 K and + 0.16 eV, i.e. endothermic, under ambient conditions (Table 2). These results indicate that re-oxidation of the surface is difficult at room temperature. The last step in Fig. 2 is the adsorption of a second CO molecule, which reacts with the protruding oxygen of O_2 and leaves as CO_2 (**V**), so that the surface can recover its initial state. The energy release for this process is large, −4.44 (−3.91) eV at T = 0 (300) K. The presence of an additional V_O on the (001) surface has only minor influence on the various intermediates (Table 2), the main effect being to slightly increase the likelihood of O_2 adsorption.

The intermediates of the COOR on Co_3O_4(110) are shown for the case of adsorption at the O1 site (Fig. 3). After CO adsorption and CO_2 formation, desorption of CO_2 to form a surface V_O has an energy release of −0.64 eV at RT, close to the analogous value on the (001) surface. At variance with the (001) surface, however, the subsequent adsorption of O_2 at the vacancy site on Co_3O_4 (110) is very favorable, being exothermic at both T = 0 K (−1.34 eV) and RT (−0.77 eV). The O_2 adsorption configuration is characterized by one of the oxygen atoms occupying the missing lattice oxygen site, while the other oxygen stretches out toward a neighboring Co(Oh) with an O—O bond length of 1.37 Å. This oxygen is very reactive; when another CO molecule arrives, CO_2 forms readily, without barrier and with an energy release of −4.28 (−3.74) eV. In the case of CO adsorption at the O2 site, CO_2 formation does not involve the CO bridging structure in Fig. 3, but is more similar to that on the (001) surface. As shown in Table 2, the energy release of step I → II is −1.43 eV at RT, compared to −0.60 eV for the O1 site, while that of CO_2 desorption (step II → III) is −0.38 eV, compared to −0.64 eV for the O1 site. Similar to the O1 site, however, also at the O2 site adsorption of O_2 is very favorable, −0.47 eV at RT, and further CO oxidation occurs with an energy release of −3.45 eV.

By comparing the free energy changes for the various steps of the COOR on Co_3O_4 (110) and (001) surfaces (Table 2), it appears that the thermodynamic limiting step of the COOR on Co_3O_4(001) is the adsorption of O_2, which becomes unfavorable around room temperature. On the Co_3O_4 (110) surface, instead, reoxidation is easy and the COOR is more likely controlled by the kinetics of CO_2 desorption, which requires a significant structural rearrangement, whereas O_2 adsorption is quite favorable.

3.2. $NiCo_2O_4$ (001)/(100) surfaces

Unlike Co_3O_4, which is a normal spinel, NCO has an inverse spinel structure, where tetrahedral (Td) sites are occupied primarily by Co^{2+} and Co^{3+} ions, and octahedral (Oh) sites by Ni^{2+}, Ni^{3+} and Co^{3+} ions [29]. NCO (001)/(100) surfaces can expose both Co(Oh) and Ni cations or only one type of metal cation; following Ref [29]., we denote $(100)_{mix}$ a surface exposing both Co(Oh) and Ni, while $(001)_{Ni}$ and $(001)_{Co}$ expose only Ni and Co(Oh), respectively.

Fig. 4 and Table 3 show the computed intermediates and free energy changes for the COOR on pristine and defected NCO(001)/(100) surfaces. Comparison of these results to those in Table 2 for Co_3O_4 shows that the free energy release for CO adsorption and CO_2 formation (step

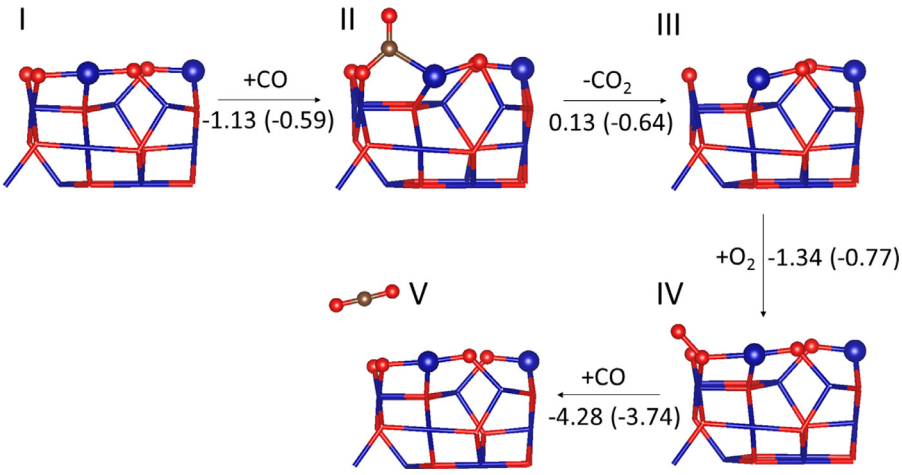


Fig. 3. Intermediates (side views) and free energy changes (in eV) for the COOR on Co_3O_4 (110), with initial CO adsorption at the O1 site. The sequence of intermediates is the same as in Fig. 2. Free energy changes are given at both T = 0 K and T = 300 K (between parentheses) with CO, O_2 and CO_2 partial pressures of 0.67 atm, 0.33 atm and 0.0004 atm, respectively.

Table 2
Computed free energy changes (in eV) for the steps of the COOR on the Co_3O_4 (001) and (110) surfaces shown in Figs. 2 and 3. The labels P/D indicate pristine and defected (1 V_O per surface cell) surface, respectively. Results are reported at both T = 0 K and T = 300 K (between parentheses) with CO and O_2 pressure of 0.67 and 0.33 atm, respectively.

	I → II	II → III	III → IV	IV → V	V → I
O1 site (001) P	−1.86(−1.32)	0.15(−0.62)	−0.42(0.16)	−4.44(−3.91)	0.04(−0.73)
O1 site (001) D	−1.75(−1.21)	0.14(−0.63)	−0.53(0.05)	−4.58(−4.05)	0.19(−0.58)
O1 site (110) P	−1.13(−0.59)	0.13(−0.64)	−1.34(−0.77)	−4.28(−3.74)	0.10(−0.68)
O2 site (110) P	−1.97(−1.43)	0.40(−0.38)	−1.05(−0.47)	−3.99(−3.45)	0.09(−0.68)

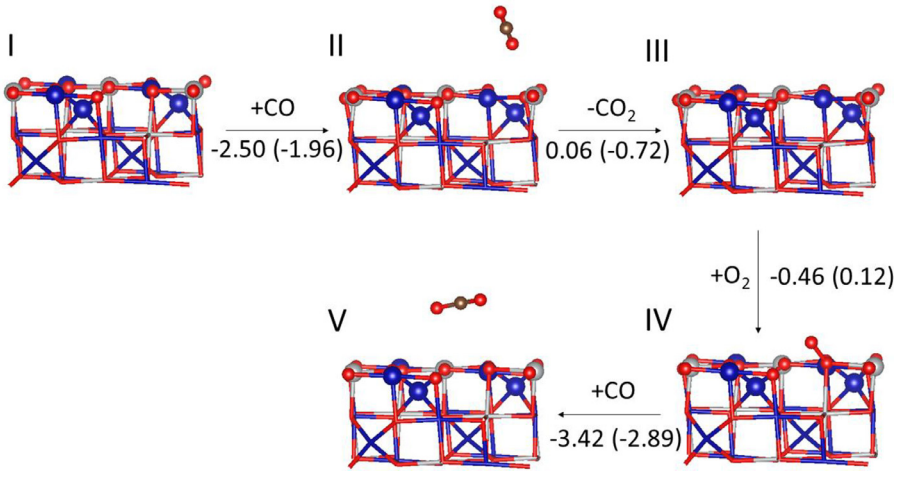


Fig. 4. Intermediates (side views) and free energy changes (in eV) for the COOR on the NCO $(100)_{mix}$ pristine surface. The sequence of intermediates is the same as in Fig. 2. Free energy changes are given at both T = 0 K and T = 300 K (between parentheses) with CO, O_2 and CO_2 partial pressures of 0.67 atm, 0.33 atm and 0.0004 atm, respectively.

Table 3
Computed free energy changes (in eV) for the various steps of the COOR on pristine (P) and defected (D) NCO (001)/(100) surfaces. Results are reported at both T = 0 K and T = 300 K (between parentheses) with CO and O_2 pressure of 0.67 and 0.33 atm, respectively.

	I → II	II → III	III → IV	IV → V	V → I
P - $(001)_{Ni}$	−2.17(−1.63)	0.28(−0.49)	−0.39(0.18)	−4.23(−3.69)	−0.03(−0.80)
D - $(001)_{Ni}$	−2.03(−1.50)	0.20(−0.57)	−0.62(−0.05)	−3.22(−2.68)	−0.85(−1.63)
D - $(001)_{Co}$	−2.17(−1.64)	0.09(−0.68)	−0.43(0.15)	−3.73(−3.19)	−0.29(−1.07)
P - $(100)_{mix}$	−2.50(−1.96)	0.06(−0.72)	−0.46(0.12)	−3.42(−2.89)	−0.21(−0.98)
D - $(100)_{mix}$	−2.10(−1.56)	0.27(−0.50)	−0.55(0.03)	−4.25(−3.71)	0.09(−0.68)

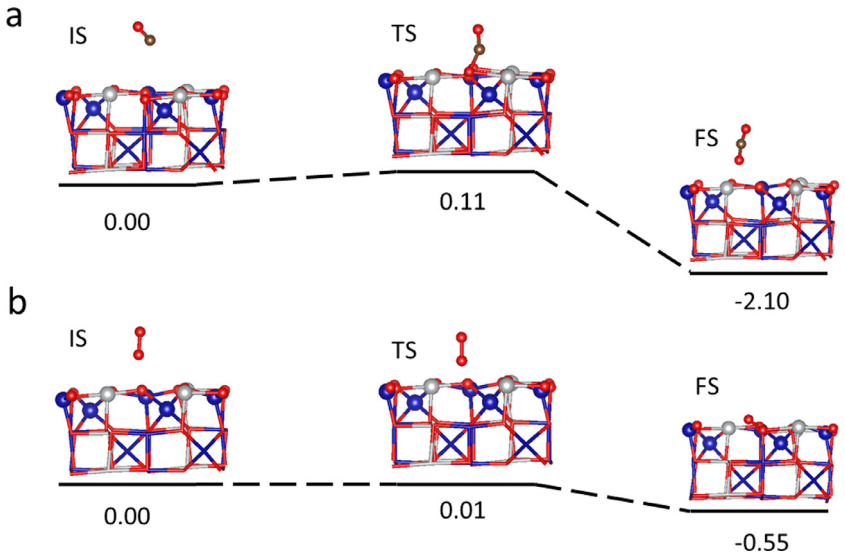


Fig. 5. Initial state (IS), transition state (TS) and final state (FS) structures and corresponding relative energies (in eV) for a, CO_2 formation (I → II) and b, O_2 adsorption (III → IV) on the D- $(100)_{mix}$ surface of NCO.

Table 4
Computed free energy changes (in eV) for the various steps of the COOR on pristine (P) and defected (D) NFO (001) surfaces. Results are reported at both T = 0 K and T = 300 K (between parentheses) with CO and O_2 pressure of 0.67 and 0.33 atm, respectively.

	I → II	II → III	III → IV	IV → V	V → I
P – (001)	−3.02(−2. 49)	0.10(−0.67)	−0.22(0.35)	−3.54(−3.00)	0.15(−0.62)
D – (001)	−2.28(−1.74)	0.10(−0.68)	−0.40(0.18)	−3.90(−3.36)	−0.06(−0.83)

I → II) on NCO is larger than on Co_3O_4. This free energy release is indeed strongly correlated with the V_O formation energy, whose value is smaller in NCO due to the presence of Ni cations in the 3rd layer [29]. On the other hand, computed O_2 adsorption energies at V_O sites (step III→IV) on reduced NCO(001)/(100) with 1/8 and ¼ ML V_O coverages are similar to those on Co_3O_4 (001), i.e. are typically positive at RT (Table 3). This suggests that reoxidation is difficult also on NCO. On the Co-terminated $NCO(001)_{Co}$ surface, in particular, V_Os tend to reconstruct [29], which prevents O_2 adsorption. Thus, the COOR cannot continue on this surface, except at high V_O concentrations, where non-reconstructed V_Os are also present. Reduced surfaces always adsorb O_2 more easily than pristine ones, as expected.

To further characterize the COOR on NCO, we studied the kinetic pathways of two key steps of the reaction, the formation of the 1st CO_2 (I → II) and O_2 adsorption (III → IV) (Fig. 5). For the 1st CO adsorption/CO_2 formation, NEB calculations give a barrier of 0.11 eV (Fig. 5(a)). The C = O bond length at the transition state is 1.12 Å, while the distances of the C atom to surface oxygen and Ni atoms are 1.35 and 2.17 Å, respectively. For O_2 adsorption on defected NCO $(100)_{mix}$, NEB calculations show a smooth potential energy surface with a very small barrier of about 0.01 eV (Fig. 5(b)). The O—O bond length at the transition state is 1.239 Å and the distance of the lower oxygen to the surface is 1.403 Å, indicating that interaction with the surface is still weak. These results suggest that the COOR on NCO is controlled by the thermodynamics of O_2 adsorption.

3.3. $NiFe_2O_4$ (001) surfaces

Among (001) surfaces of Co_3O_4, NCO and NFO, the interaction of CO with surface metal cations is weakest on NFO(001). We found indeed an average CO adsorption energy of −0.11 and −0.42 eV on Fe and Ni sites of NFO(001), to be compared to −0.31 and −0.88 eV on Ni and Co sites of NCO (100)/(001), and −0.92 eV on Co sites of Co_3O_4(001). On the other hand, NFO(001) has the lowest computed V_O formation energy among these materials[23, 29]. This has important consequences on the COOR. As shown in Table 4 and Fig. S2 (SI), the free energy change for the step of CO_2 formation (I → II) of the COOR is significantly more favorable on NFO(001) than on Co_3O_4 (100) and NCO (001)/(100) surfaces. On the other hand, the step of O_2 adsorption at a V_O (III → IV) is less favorable on NFO(001) in comparison to Co_3O_4(100) and NCO (001)/(100), which may explain why NFO is a less efficient COOR catalyst than Co_3O_4 and NCO.

As the NFO (001) surface tends to be hydroxylated in humid environment [23], we also investigated the influence of surface hydroxyls on CO adsorption. We used a structural model where all O1 sites are changed into OH groups, which represents the stable surface in a wide range of temperatures [23]. We found that besides blocking the O1 sites, the hydroxyls also hinder the interaction of CO with the bare (i.e. non-hydroxylated) O2 sites: instead of immediately reacting with these sites, as found on the non-hyroxylated surface, CO tends to remain at least 2.5 Å away from the surface, suggesting the presence of an activation barrrier. Altogether, this indicates only weak interaction between the hydroxylated surface and gaseous CO, which could also contribute to the low COOR activity of NFO.

4. Conclusion

We have used DFT + U calculations to identify similarities and differences in the CO oxidation activities of (001) and (110) surfaces of Co_3O_4, $NiCo_2O_4$ and $NiFe_2O_4$, three spinel oxides all made of earth-abundant elements. Our results indicate that the (001) surfaces of these materials have all the same mechanism for the COOR. CO molecules adsorb strongly at lattice oxygen sites and form CO_2 readily, while re-oxidation through O_2 adsorption at V_O sites is thermodynamically unfavorable at room temperature or higher. On the more reactive Co_3O_4 (110) surface, instead, the re-oxidation is relatively easy and the rate-limiting step of the COOR is the formation of CO_2 from adsorbed CO. At lower temperature, competing molecules like water can easily adsorb and dissociate at V_O sites, thus preventing the interaction of O_2 with the sites and blocking the COOR activity. This effect is particularly important for surfaces with low V_O formation energy, notably $NiFe_2O_4$ (001), consistent with the low observed COOR activity of this surface.

Acknowledgments

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